

Aerodynamic Study of a Low-Speed Fan in Distorted Inlet Conditions

(Bibliography report)

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I. CONTEXT

Turbo-engines are often demanded to work in off-design conditions which can deteriorate performances and stability of the compression system. Correct understanding and prediction of the behaviour of turbomachines in such conditions is not an easy task. Engineers traditionally face this problem with three different methods:

- Low order analytical models that capture the main physics of the problem and rapidly give approximated results;
- High fidelity numerical simulations which attain high levels of accuracy but require long computational time;
- Direct observations and experimental measurements made on compressor and system test sessions.

In system testing is far more important to collect quickly the data for results comparison and trends interpretation. For this purpose, and for preliminary system design, simplified models are the best analysis tool. In view of this, the aim of this work is to develop a simple model, using Matlab and Excel, for the prediction of low speed fan performances in off-design conditions. This problem is interesting for fan applications where interactions with the system create a distorted flow at the machine inlet. This is the case of avionics ventilation fans or engines with boundary layer ingestion (BLI) [1].

II. PROBLEM STATEMENT

There are three main types of flow distortions at compressor inlet: total pressure distortion, total temperature distortion, and inlet swirl. The term 'distortion' means a non-uniform distribution of these quantities along the radial and/or circumferential direction on the compressor annulus. They are caused, for example, by intake flow separation during manoeuvres or by boundary layer ingestion. An inlet swirl consists in the presence of a circumferential speed component in the inlet flow which changes the incidence on compressor blades. Inlet swirl can be induced by an IGV component placed forward the rotor or by S-duct intakes. At constant throttle ($M_1 = \text{const}$) and geometry ($S_1 = \text{const}$), total pressure and total temperature variations at compressor inlet cause mass flow rate variations that result in axial speed variations on the annulus:

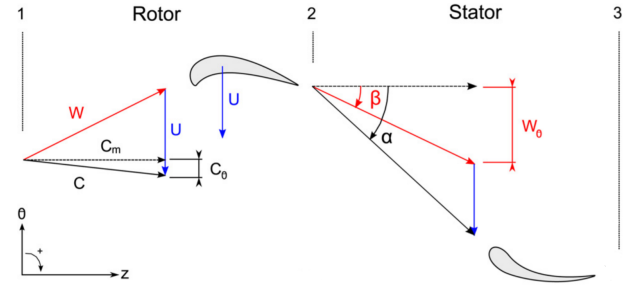


Fig. 1. Conventions and references in velocity triangle for a conventional fan stage [2].

$$\dot{m}_1 = \frac{p_t S_1}{\sqrt{RT_t}} f(M_1) = \rho C_m S_1^1 \quad (1)$$

Therefore, it is possible to consider the effect of these two types of distortions through the calculation of the local flow coefficient ϕ in each point of the compressor inlet. This parameter is in fact directly related to the local work exchange, expressed by the local loading coefficient ψ . Under appropriate hypothesis the $\phi - \psi$ relation depends only on the local blade geometry as shown in [2] and [3].

The simplified model traditionally used to analyse the performance of a compressor with distorted inlet flow is the Parallel Compressor Model, as reported in [4]. It consists in ideally subdividing the compressor in segments along the circumferential direction and consider each of them as a single compressor submitted to uniform inlet conditions and working independently from the others. This simple model is well adapted for simple patterns of circumferential distortions, but becomes rapidly inaccurate with more complex configurations. For this reason, many extensions have been proposed to obtain more accurate results with complex distortion patterns [5]. For example, Reid [6] and Korn [7] introduced the fundamental concept of critical angle of circumferential distortion extent. The lag in the aerodynamic response of blades traversing the distortion sector while turning may reduce the impact of the

¹Mass conservation in the intake duct

distortion when its extension is inferior to the critical angle. Considering this effect, a far better approximation of test results was obtained. Later Cousins [8] explains the concept of critical angle in terms of flow physics, defining the rotor response time as the time necessary for a particle to go from blade leading edge to blade passage throat. The account of dynamic effects may have a major role in capturing the real problem physics and increases model accuracy, as seen in studies made by Mardoc and Van Deusen in [9]. Other important phenomena are the radial and circumferential flow redistribution, due to pressure gradients different from the non-distorted case, and also the distortion angular displacement, due to rotor stage rotation. They are the primary cause of the distortion attenuation through the stages. Circumferential redistributions can be modelled with good approximation using an orifice flow analogy in the gap between rotor and stator as stated in [10]. However, modelling radial redistributions is still problematic because they are strictly related to ‘radial equilibrium’ and radial loading distribution, modified by the presence of the distortion. In [12] Mazzawy shows how to evaluate the distortion angular displacement due to its passage in the rotor stage.

Another critical point is simulating blade action maintaining simplicity. The actuator disk model results too reductive due to its limited geometric representativity, not sufficiently accurate in presence of inlet distortions. The traditional approach uses compressor characteristic maps, see [4] associated to the specific machine, but often results too simplistic because local behaviour can be slightly different for high aspect ratio blades like in fans. Another possibility is the implementation of a mean line code, associated to blade geometry, which directly calculates local characteristics from local velocity triangles and inlet conditions. In this way swirl response and loading parameters are better predicted and a larger amount of flow data is available [4]. Another important approach found in literature is the ‘body-force’ technique. An example is the DYNTECC code [15] which uses mass, momentum and energy balance equations associated with turbomachinery source terms to consider blade action. This technique has the advantage to intrinsically capture three-dimensionality, real geometry and unsteadiness, but results more complex and costly than the parallel compressor model. On the other hand, many off-design phenomena like dynamic blade row response and flow separation must be modelled and experimentally calibrated using a parallel compressor technique.

III. FIRST DEVELOPMENTS AND FUTURE WORK

My simplified model aims to simulate the functioning of an axial low speed fan in wind-milling regime [2] with a distorted inlet flow. Fan stage is composed of a rotor, a stator and possibly an IGV component forward the rotor. In my model I decided to divide the fan stage in small cellules, using an approach similar to the parallel compressor method, aiming to obtain a good balance between accuracy and simplicity. The choice of using local $\phi - \psi$ relations for rotor blade row actions has been made to accurately characterise the

radial differences in geometry. The outlet deviation model chosen is the Carter’s Law [14], but it is necessary to find a proper correction for blades working in turbine mode. Indeed, from experimental measures made in wind-milling regime, the deviation angle β_2 at rotor outlet results bigger than Carter’s Law predictions for blades in turbine mode [2]. The characterisation of inlet distortion is made through a non-uniform pattern of ϕ at inlet and the swirl effect is considered through an $\alpha_1 \neq 0$ inlet distribution. To account for the contraction of the meridional channel (not purely axial configuration) a corrected expression for the $\phi - \psi$ relation has been determined, considering the differences of blade velocity U between inlet and exit rotor plan. An important but generally neglected phenomenon is viscosity, that I would try to simulate in a simple way through blockage of wall boundary layers and total pressure losses. Other critical implementations are the radial pressure equilibrium at rotor outlet and the circumferential flow redistributions at rotor-stator gap. These effects are strictly related to the distortion attenuation through blade rows. For now, my analysis is limited to rotor stage, but I envisage to implement the presence of an IGV component and the stator stage.

Results comparisons for model calibration and validation are made with existing experimental measurements taken from LP4 reversible fan and with the results of high fidelity CFD simulations. The model should be firstly validated for working conditions near the design point and with uniform inlet conditions. The further step would be the validation at wind-milling regime, firstly without an inlet distortion and then with inhomogeneous inlet flow patterns. Once the model will be validated, it will be possible to rapidly predict the response to specific distortion patterns for various fan configurations and to explore the effect of main geometry parameters variation, in order to propose solutions to actual off-design issues.

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